ChatGPT

The 21 cm Hydrogen Line in the Cosmic Dark Ages

Physical Basis of the 21 cm Transition

The 21 cm line arises from the hyperfine splitting of neutral hydrogen's ground state: parallel (triplet) vs anti-parallel (singlet) electron–proton spin configurations. The energy difference is ΔE≈5.9×10^−6 eV (wavelength λ=21.1 cm, frequency ν=1420 MHz) [1]. Transitions between these levels are extremely slow: the spontaneous decay (triplet→singlet) rate is A_10 ≈ 2.85×10^−15 s^−1 (lifetime ~10^7 years) [2]. In practice, however, the vast abundance of hydrogen and extra excitation mechanisms make the line observable. The hydrogen "spin temperature" T_S is defined by the relative populations of the two states (n_1/n_0 = (g_1/g_0) exp(–T/*T_S), with T* ≈0.068 K) [1]. Collisions and scattering of ambient photons (e.g. Lyman-α or CMB) drive T_S toward either the gas kinetic temperature T_K or the CMB temperature T_CMB, depending on redshift and environment.

Collisions dominate during the Dark Ages, so T_S ≈ T_K. At high z immediately after recombination, gas and CMB were tightly coupled and T_K ≈ T_CMB. But as the Universe expands, the gas cools adiabatically (T_K ∝ (1+z)^2) faster than the radiation (T_CMB ∝ (1+z)). Collisions initially keep T_S ≈ T_K < T_CMB, causing 21 cm absorption. At lower z the gas becomes too diffuse for collisions to maintain the coupling, and T_S tends back toward T_CMB (weakening the signal) until the first stars form. These atomic physics and radiative transfer basics determine the 21 cm brightness relative to the CMB [1] [2].

The Cosmic Dark Ages (z≈30–200)

The *cosmic Dark Ages* span roughly redshifts z≈30–200, after recombination (z≈1100) and before the first stars. During this era, the Universe is nearly fully neutral and optically thin, and the only radiation field is the CMB and any residual Lyman-α from recombination. As Pritchard & Loeb explain, after baryons thermally decouple from the CMB, the gas cools and its 21 cm line enters absorption against the CMB [3]. Because no astrophysical sources yet exist, Lyα coupling is absent and collisions (with H, e^−, He) are the only coupling mechanism. As a result, the spin temperature follows the gas temperature: T_S ≈ T_K < T_CMB, producing a weak 21 cm absorption trough. Eventually (around z~30–50, depending on density) collisions become too rare and T_S → T_CMB; at that point the absorption signal fades away. In summary, the Dark Ages 21 cm signal is a broad, shallow absorption feature spanning z~30–100 [3] [4].

The 21 cm Signal as a Cosmological Probe

The 21 cm transition provides a **3D map of neutral hydrogen** across cosmic time, linking the CMB epoch and the later universe. The observable quantity is the *differential brightness temperature* δT_b (relative to the CMB), which for diffuse IGM can be approximated as:

$$\delta T_b \approx 27\,\mathrm{mK}\;\; x_{\mathrm{HI}}\,(1+\delta)\left(\frac{\Omega_b h^2}{0.023}\right)\left[\left(\frac{0.15}{\Omega_m h^2}\right)\left(\frac{1+z}{10}\right)\right]^{1/2}\left(1-\frac{T_\gamma}{T_S}\right),$$

where x_{HI} is the neutral fraction, δ the matter overdensity, Ω's are cosmological parameters, and $T_\gamma = T_{\rm CMB}(z)$ [5] [6]. At high z in the Dark Ages, $x_{HI} \approx 1$ and δ≈0 on large scales, so the signal depends mainly on (1–T_γ/T_S). If T_S<T_CMB (absorption) δT_b<0; if T_S>T_CMB (emission) δT_b>0.

21 cm observations include both the **sky-averaged (global)** signal and its spatial **fluctuations**. The global signal (Figure 1, top panel) marks the timing of cosmic milestones (first star formation, X-ray heating, reionization) [7]. The fluctuating signal (Figure 1, bottom) traces the 3D matter density and ionization field [7]. In the Dark Ages (no stars), fluctuations are purely cosmological (density+velocity) and the brightness field is nearly Gaussian. Detecting these fluctuations would allow precision tests of cosmology (e.g. the primordial power spectrum and neutrino mass). As shown in Figure 1, the global Dark Ages absorption is predicted to be only ~–50 mK around z~80 [4], far weaker than the deep cosmic-dawn trough (tens to hundreds of mK).

Figure: 21 cm brightness temperature across cosmic time (mK). The **top** panel shows the global (sky-averaged) signal vs. redshift (z), covering the Dark Ages, Cosmic Dawn, and Epoch of Reionization. The **bottom** panel shows a simulated lightcone of spatial fluctuations over the same period (vertical axis ~ comoving position, horizontal ~ cosmic time). In this standard model: during the Dark Ages (shaded) neutral hydrogen produces only a weak absorption (~–50 mK) [4], followed by a stronger absorption trough during Cosmic Dawn (z~15–20) when first stars turn on, and then an emission peak as X-ray heating warms the IGM, ending in zero signal once reionization completes.

Theoretical Predictions (Signal Strength)

Analytical and numerical models predict the global 21 cm signal and power spectrum given assumed astrophysics. In standard ΛCDM, the **global Dark Ages absorption** peaks at ~–50 mK at z~70–90 [4]. The **Cosmic Dawn trough** (with first Lyα photons) might be ~–100 to –200 mK deep around z~15–20. These amplitudes scale with the unknown early heating rate; exotic physics (e.g. dark matter–baryon scattering) could deepen them. The **fluctuation power** during the Dark Ages is expected to be very small (mK-scale) and largely cosmological [8]; during Cosmic Dawn/EoR it is higher due to patchy heating and ionization.

Equation (1) above shows δT_b ∝ $\sqrt{}$(1+z) and vanishes if T_S=T_CMB. In practice, theoretical models yield brightness fluctuations of order a few mK at z~10–20 and global features of tens of mK [5] [4]. For example, Giri et al. (2023) use Equation (1) and simulations to find δT_b ~20–30 mK in emission at late reionization, with dark-age absorption $\lesssim$–50 mK [5] [4]. These predictions serve as targets for experiments.

Observational Challenges and Foregrounds

Detecting the Dark Ages 21 cm signal is extremely challenging. First, **frequency**: z~30–200 corresponds to ν $\lesssim$50 MHz (wavelength $\gtrsim$6 m). Earth's ionosphere becomes opaque below $\lesssim$10–30 MHz, and causes refraction/distortions even above that. The long wavelengths demand huge antennas (or arrays) and suffer strong terrestrial RFI. Second, **foregrounds**: the sky at ν<200 MHz is dominated by Galactic synchrotron (brightness hundreds to thousands of kelvin) and extragalactic radio sources [9]. For example, the cosmic signal (tens of mK) is ≈10^5–10^6 times fainter than the Milky Way emission [9]. Removing these smooth-spectrum foregrounds requires exquisite calibration and subtraction. Finally, instrumental systematics

(beam patterns, reflections, polarization leakage) must be controlled to part-in-10^5 accuracy, or else spurious spectral structure will mimic or mask the 21 cm signature [10] [11].

In practice, these issues mean that cosmological 21 cm observations must carefully avoid RFI (remote sites or space) and model/mitigate smooth foregrounds. [9] [12]. Even so, current experiments can generally place **upper limits** on the 21 cm power spectrum, but no unambiguous detection yet (see next section).

Major Experiments and Observatories

Key efforts are targeting the cosmic dawn and Dark Ages 21 cm signal:

- **Global (single-antenna) experiments:** e.g. **EDGES** (MIT/ASU) in Australia [13], **LEDA** (Caltech/JPL in New Mexico) [14], **SARAS** (India) [15], **PRATUSH** (India, planned lunar orbiter) [16], and proposed space missions like **DARE**. These aim to measure the sky-averaged spectrum. EDGES famously reported an absorption trough at 78 MHz (z≈17) in 2018 [13], much stronger than expected, though its validity remains debated [17] [18].

- **Interferometric arrays:** These seek the power spectrum or images of 21 cm fluctuations. Current arrays include the **LOw-Frequency ARray (LOFAR)** in Europe, the **Murchison Widefield Array (MWA)** and **Hydrogen Epoch of Reionization Array (HERA)** in Western Australia, and the **Precision Array for Probing the EoR (PAPER)** in South Africa. These have placed stringent upper limits on the 21 cm power spectrum at z≈6–12 [19]. The future **Square Kilometre Array (SKA-Low)**, under construction in Australia, will have vastly greater sensitivity, aiming for detailed imaging and detection of the signal across z~6–30 [12] [20]. Table 1 summarizes representative projects:

Experiment	Type	Frequency Range	Target Epoch	Notes
EDGES	Global	50–200 MHz	z~6–30	Detected tentative absorption at z≈17 [13]
SARAS	Global	40–200 MHz	z~6–30	No detection of EDGES-like feature [18]
LEDA	Global/ Array	30–88 MHz	z≈15–30	Station at LWA, foreground-calibrated spectrum [14]
LOFAR (NL/ EU)	Array	120–240 MHz	z~6–11	Upper limits on power spectrum [19]
MWA (Aus)	Array	80–300 MHz	z~6–12	Upper limits and imaging pathfinder [19]
HERA (SA)	Array	50–250 MHz	z~6–20	In commissioning, targets power spectrum (not yet detected) [19]
SKA-Low	Array	50–350 MHz	z~6–30	Future large array, will image 21 cm sky [12]

Cosmological and Astrophysical Implications

A real 21 cm measurement would revolutionize our understanding of the early Universe. It directly probes the thermal history (gas temperature T_K) and ionization fraction of the IGM. For example, the global 21 cm troughs mark when the first stars turned on and when X-ray sources heated the gas. The end of detectable 21 cm signal pinpoints the completion of reionization. By mapping fluctuations, one can constrain the growth of structure, neutrino masses (via small-scale power), and primordial non-Gaussianity.

Importantly, the signal is sensitive to new physics. The unexpected EDGES result (if taken at face value) spurred ideas such as dark matter–baryon scattering (to cool the gas) or excess radio backgrounds [17]. In general, 21 cm data can place limits on dark matter annihilation/decay (which would heat the gas), interactions that modify the gas temperature, or exotic early energy injection. The Dark Ages 21 cm signal specifically is a clean probe of fundamental cosmology because it is prior to astrophysical complications [8] [21].

In summary, the 21 cm line from neutral hydrogen during z~30–200 offers a unique window into the "last frontier" of the cosmic timeline [21] [22]. Though no detection has yet been made from this era, ongoing and planned experiments (ground-based and space-based) strive to reveal it. A robust 21 cm measurement would constrain the thermal and ionization history of the Universe, the properties of the first stars and black holes, and even new particle physics or dark matter models.

Key References: Representative sources include Giri et al. (2023) [22] [5], Pritchard & Loeb (2012) [3], Fialkov et al. (2023, Phil. Trans. R. Soc. A) [21] [8], and project resources such as the EDGES website [13] and the SKA Observatory science pages [20], among others cited above. These cover the physics of the 21 cm line, modeling of the Dark Ages signal, and experimental efforts to detect it.

---

[1] [3] [1109.6012] 21-cm cosmology in the 21st Century
https://ar5iv.labs.arxiv.org/html/1109.6012

[2] The Epoch of Reionization - Saleem Zaroubi
https://ned.ipac.caltech.edu/level5/March14/Zaroubi/Zaroubi4.html

[4] [7] [8] [2311.05366] Cosmic mysteries and the hydrogen 21-cm line: bridging the gap with lunar observations
https://ar5iv.labs.arxiv.org/html/2311.05366

[5] [6] [12] [19] [22] Suppressing variance in 21 cm signal simulations during reionization
https://www.aanda.org/articles/aa/pdf/2023/01/aa44986-22.pdf

[9] [11] [15] [16] [17] [18] Signals from the depths of the cosmos
https://www.nature.com/articles/d44151-023-00183-0?error=cookies_not_supported&code=8948f9aa-835c-43a6-96bf-2f65a8592bdf

[10] [13] EDGES: Experiment to Detect the Global EoR Signature - MIT Haystack Observatory
https://www.haystack.mit.edu/astronomy/astronomy-projects/edges-experiment-to-detect-the-global-eor-signature/

[14] Large Aperture Experiment to Detect the Dark Ages - Wikipedia
https://en.wikipedia.org/wiki/Large_Aperture_Experiment_to_Detect_the_Dark_Ages

20 Probing the Cosmic Dawn | SKAO
https://www.skao.int/en/explore/science-goals/164/probing-cosmic-dawn

21 [2311.05366] Cosmic mysteries and the hydrogen 21-cm line: bridging the gap with lunar observations
https://arxiv.org/abs/2311.05366